

JIHANG LI

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SUMMARY

- Master of Computer Science student conducting 3D cell-lineage reconstruction research (CellUniverse), designing Monte Carlo optimization and PCA-based cell-division detection algorithms in C++ for biological imaging.
- Software engineer at startup LaunchPath, building an AI-powered sales-outreach email platform end-to-end — Spring Boot and React, RAG-grounded email generation with pgvector, and autonomous LLM agent workflows.
- Quickly learn and master new technologies; successful working in both team and self-directed settings.

EDUCATION

UNIVERSITY OF CALIFORNIA, IRVINE, Irvine, California
Master of Computer Science, Expected December 2026, GPA: 3.97/4.00

UNIVERSITY OF CALIFORNIA, SANTA CRUZ, Santa Cruz, California
B.S., Computer Science, 09/2021 – 12/2024, GPA: 3.67/4.00

TECHNICAL SKILLS

Languages:	Java, Python, C/C++, JavaScript, HTML/CSS
Frameworks:	Node.js, Django, REST API
Developer Tools:	Git, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Jupyter Notebook
Libraries:	Pandas, NumPy, Matplotlib, PyTorch, TensorFlow

EXPERIENCE

LAUNCHPATH, Remote, California

Software Engineer, LaunchMail — AI sales-outreach email platform

Jun 2026 - Present

- Shipped LaunchMail, an AI sales-email platform (Spring Boot, React/TypeScript, PostgreSQL) with 110+ REST endpoints.
- Implemented RAG-grounded AI email drafting with pgvector embeddings, unified across 5 LLM providers.
- Built autonomous outreach agents and a drag-and-drop workflow engine on a webhook-resumed Postgres state machine.
- Engineered deliverability safeguards: sender rotation, randomized throttling, daily caps, and a 220-term spam linter.
- Ensured reliability with 242 integration tests (Testcontainers, WireMock) across 33 Flyway schema migrations.

UNIVERSITY OF CALIFORNIA, SANTA CRUZ, Santa Cruz, California

Undergraduate Researcher, AIEA Lab, Research under Dr. Leilani Gilpin

Jan 2024 - Jun 2025

- Research Gaussian splatting techniques to reconstruct dangerous scenes, enhancing vehicle response to hazard situations.
- Conducted object detection research to improve recognition in complex urban areas, using machine learning algorithms.
- Utilized Python, TensorFlow, and PyTorch for experiments, data analysis, and model development.
- Collaborated with a multidisciplinary team to advance state-of-the-art research in autonomous driving technology.
- Improved Gaussian-splatting rendering efficiency by 30%, accelerating playback of reconstructed hazard scenes for analysis.

PROJECTS

CellUniverse 3D Cell Tracking • Software Engineer • Algorithm • C++ • [Github](#)

Feb 2026 - Present

MCS Keystone & Capstone research lead by Dr. Wayne Hayes's focusing on 3D cell lineage reconstruction.

- Replaced voxel-matrix rendering with analytical inverse-rotation ellipsoid testing, reducing memory and fixing artifacts.
- Built Monte Carlo optimizer fitting 3D oblate spheroids to confocal microscopy data via L2 cost minimization.
- Developed PCA-based cell division detection clustering real-image bright pixels to find split axes and daughter positions.
- Implemented per-axis normalization and neighbor-exclusion filtering to prevent PCA contamination from adjacent cells.
- Designed two-phase optimization separating cell fitting from split detection to prevent cascading false divisions.
- Contributing to a research manuscript on automated 3D cell lineage tracking across biological imaging datasets.

Doom RL Agent • Machine Learning Engineer • Reinforcement Learning • Python • [Github](#)

Sep 2025 - Dec 2025

Collaborative RL project training PPO agents to play the 1993 Doom via VizDoom scenarios

- Implemented PPO agents via Stable-Baselines3 targeting three VizDoom scenarios with increasing enemy complexity.
- Designed a reward shaping system with survival bonuses, death penalties, and kill rewards to guide exploration.
- Applied running-statistics reward normalization to stabilize gradients and accelerate policy convergence.
- Tuned PPO hyperparameters (learning rate decay, batch size, entropy coefficient) improving survival from ~38 to 300+ steps.
- Containerized training with a GPU-accelerated Docker environment using CUDA 11.8 and docker-compose.